Solar Panel Detection | Median Income Analysis

Write-up:

Methods

I used the SAM3 solar panel predictions for Oahu (oahu_solar_dedup_simple.geojson), which contained 88,663 polygons. I repaired invalid geometries and reprojected the data to UTM Zone 4N (EPSG:32604) so polygon areas would be in square meters to be more compatible with our analysis. With some help from Claude, I was able to successfully convert the polygons into square meters. Median household income by census tract came from the 2022 American Community Survey 5-year estimates (table B19013_001), pulled with the tidycensus package for Honolulu County. Unknowingly, Honolulu County also includes the Northwestern Hawaiian Islands. I was able to zoom in to Oahu only using st_bbox, which made it easier to look at Oahu only. I then only kept tracts whose centroids fell within an Oahu bounding box, leaving 328 tracts.

Polygon sizes were heavily right-skewed (median 8691 m², maximum 511053 m²). I looked at essentially the percentiles as well as the largest polygons on satellite imagery to decide how to separate household-scale from commercial-scale installations; as well as used my own judgement to make my best guess. I also grouped polygons within 10m of each other into clusters to check whether larger installations are split into many smaller polygons. I removed polygons larger than 500 m². The histogram of polygon areas shows most detections fall between roughly 5 and 100 m², consistent with rooftops, with much larger polygons beyond that.

I assigned each remaining polygon to a census tract by its centroid so no polygon was counted twice, then summed the area for each tract. Tracts with no polygons were given a value of zero. I joined the totals to the income data and dropped tracts with no income estimate, leaving 311 tracts. I made a scatter plot of solar area (m²) against median income, with each point representing one tract, and calculated two correlation coefficients with p-values. I chose to include the Spearman correlation test,

which is based on ranks and less affected by extreme values. I thought this would be insightful due to the heavy tract-level solar area that was right-skewed.

Results

Solar area was unevenly distributed across Oahu. Per-tract solar area ranged from 0 to 134103 m² (median 7804 m²), and 1 tract had no detected solar. Median household income ranged from $19,524 to $216,103 (median $102,708).

The scatter plot shows a slight positive relationship between median income and solar area which is displayed in question 10. The initial correlation was r = 0.3033261 (p = 4.849e-08), and the Spearman correlation was ρ = 0.5437899 (p = 2.2e-16). This means higher-income tracts tend to have more solar infrastructure. Both relationships are statistically significant at the 0.05 level. To help me further identify the difference, I prompted claude to help me understand the biggest difference between the two correlation coefficients: The Spearman coefficient was noticeably larger than the initial test, which suggests that a few tracts with very high solar area weaken the fit even though higher-income tracts consistently rank higher in solar area overall. Income explains only about 9% of the variation in solar area ($r^2 = 0.09$), so most of the variation between tracts comes from other factors.

Solar farms were not usually a single large polygon. Many of the largest clusters were made up of 65 to 148 polygons (for example, the largest cluster had 136 polygons totaling about 119,600 m²), which means a size cutoff on individual polygons alone leaves pieces of commercial installations in the data.